Distributed and Enterprise Software Development (DESD) UFCFTR-30-3

Introduction

Module Team

Module Leader(s) – Khoa Phung, Dilshan Jayatilake

Module Team:

- Khoa Phung (Co-module Leader): khoa.phung@uwe.ac.uk
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Module Delivery

- **Delivery Mode:**
  - 1-hour lecture + 2-hour practice per week
- **Practices:**
  - Agile Development (Scrum)
  - Group work – Sprint-based Development
- **Autumn Term (TB1)** covers:
  - Concepts of Services/Micro-Services
  - Concepts of Business Modelling
  - Modelling Business Concepts & Automation with BPMN
- **Spring Term (TB2)** focuses on:
  - Containerisation and Deployment
  - Development of Enterprise Systems & Services using Django/Django Rest Framework and Docker

Timetable

Full Academic Year

7-18

Day	Start	End	Activity	Module	Type	Room	Staff	Title
Monday	13:00	15:00	UFCFTR-30-3 TB1 CP_oc/01	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Wei, Kun;Liu, Yan	
Monday	15:00	16:00	UFCFTR-30-3 TB1 L_oc/01	Distributed and Enterprise Software Development	Lecture (on campus)	2B025 FR	Jayatilake, Dilshan;Phung, Khoa	
Monday	17:00	19:00	UFCFTR-30-3 TB1 CP_oc/02	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Phung, Khoa;Wei, Kun;Jayatilake, Dilshan	
Tuesday	13:00	15:00	UFCFTR-30-3 TB1 CP_oc/05	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Case, Desmond;Hosseinabady, Mohammad	
Wednesday	9:00	11:00	UFCFTR-30-3 TB1 CP_oc/04	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Rhodes, Mark;Case, Desmond	
Thursday	13:00	15:00	UFCFTR-30-3 TB1 CP_oc/03	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Javed, Barkha;Liu, Yan	

24-31, 34-37

Day	Start	End	Activity	Module	Type	Room	Staff	Title
Monday	13:00	15:00	UFCFTR-30-3 TB2 CP_oc/01	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Liu, Yan;Wei, Kun	
Monday	15:00	16:00	UFCFTR-30-3 TB2 L_oc/01	Distributed and Enterprise Software Development	Lecture (on campus)	2D067 FR	Jayatilake, Dilshan;Phung, Khoa	
Monday	18:00	20:00	UFCFTR-30-3 TB2 CP_oc/03	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Liu, Yan;Case, Desmond	
Tuesday	18:00	20:00	UFCFTR-30-3 TB2 CP_oc/02	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Javed, Barkha;Wei, Kun	
Thursday	14:00	16:00	UFCFTR-30-3 TB2 CP_oc/05	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Rhodes, Mark;Phung, Khoa	
Thursday	16:00	18:00	UFCFTR-30-3 TB2 CP_oc/04	Distributed and Enterprise Software Development	Computer practical (on campus)	3Q044 FR	Jayatilake, Dilshan;Case, Desmond	

Module Learning Outcomes (MOs)

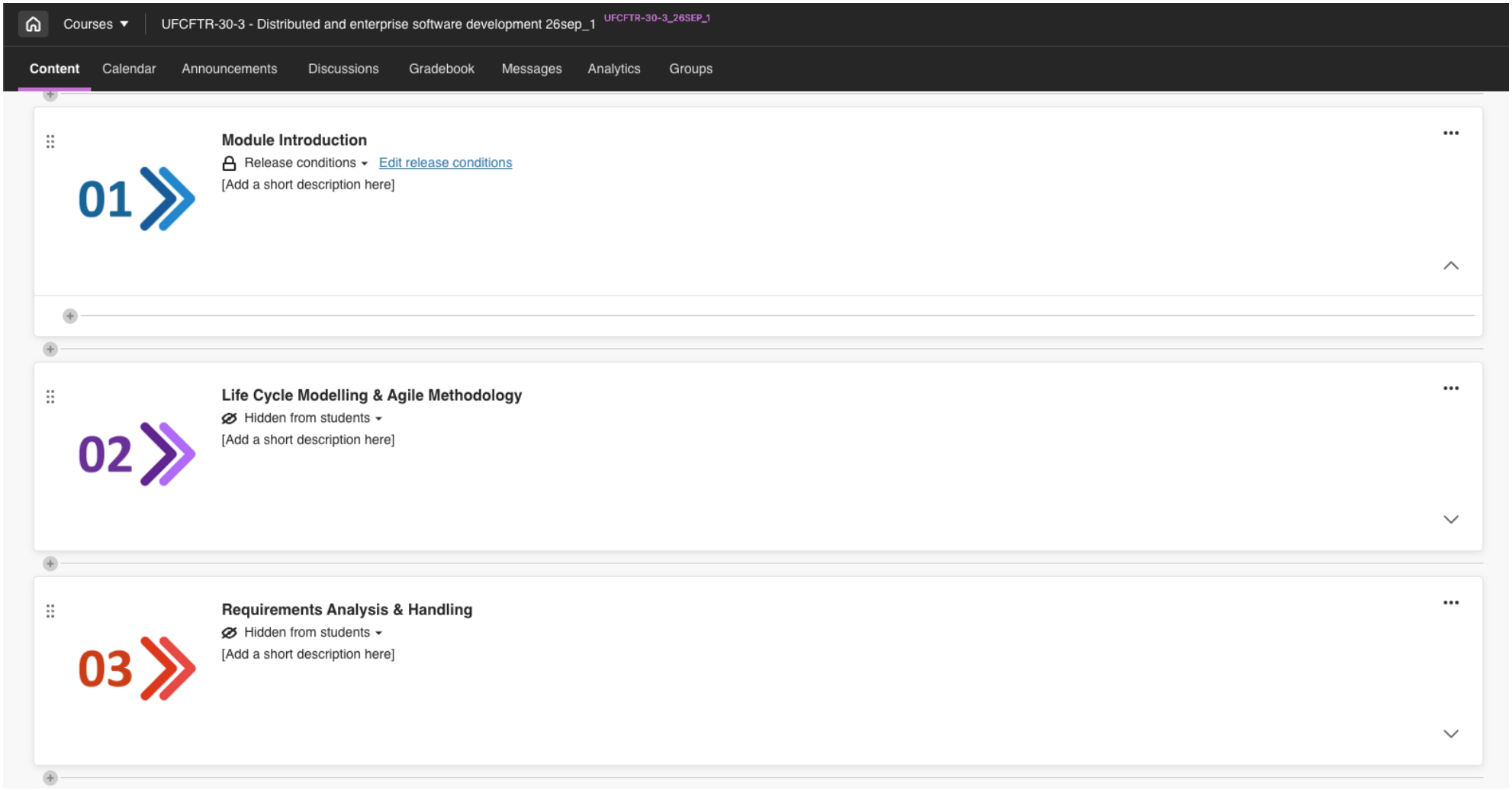
Describe	Describe and evaluate the concepts and paradigms of modern enterprise systems with particular focus on architectures, components, interfaces and services.
Evaluate	Evaluate the legal, ethical, social and professional implications in development life-cycle of large scale software systems.
Apply	Apply a current industrial project management approach when undertaking a large-scale software development project.
Identify	Identify security issues in distributed or enterprise level software systems in order to implement preventive measures.

Module Information

Please See Blackboard for

Module Information

Blackboard Learning Materials



Reading List

What to cover? (Autumn Term – TB1)

- Concepts of Distributed Systems
- Modelling Enterprise Systems, which consists of:
  - *Agile* Methodology
  - *Requirements H*andling
  - Business Analysis & Modelling
  - *Business Process Model Notation (BPMN) – Introduction & Implementation*
  - *Component and Service* Design
- Introduction to Development and Design patterns
  - MVC paradigm in development
    - "**Django**" inherently applies MVC

Teaching & Learning Tools (Autumn Term – TB1)

- The tools to be used alongside BB
  - ***BPMN & Camunda***
    - *How to design business process model?*
    - *How to automate business processes?*
  - ***Python*** *programming/scripting language*
  - ***GitHub/GitLab*** *as storage and version control tool*

What to cover? (Spring Term – TB2)

- Development with Distributed Computing
  - **Containerisation**
    - "Docker" for container development
  - **Distributed development & deployment**
    - "Multi-container" for local deployment & deployment
- A Web application as an Enterprise System, which consists of:
  - **Advance API Development** concepts
  - **Components** developed in house
  - Components recycled
  - Use of **persistent data** (DBs, e.g. Postgres, MySQL)
  - Use of **services** (web services, micro services)
  - **Security** (Authentication / Authorisation)

Teaching & Learning Tools

- Following agile methodology in project management
  - Scrum-like agile management
    - Sprint-wise practice and delivery
    - Customer is in loop (**customer = tutor**)
    - Three-week Sprints
      - *Sprint Review to be provided after each sprint*
  - Use of commonly used tools
    - Version control software such as **GitHub/GitLab**,
    - Professional Project Management tools such as *Jira, Microsoft Project* to be used
  - **No reports required!**

Lectures

- One-hour lecture sessions are planned to introduce a number of most relevant concepts, components and tools in the context of Distributed and Enterprise Systems as indicated in the Schedule provided before.
- It is timetabled in **2B025 (TB1) and 2D067 (TB2)**:
  - On Monday at 15.00

Practicals (Autumn Term – TB1)

- Timetabled in 5 Sessions:
  - In room **3Q44**, as indicated in your personal timetable
- A series of development activities are planned for practical sessions
  - **BPMN / Camunda** will be used extensively for modelling & automating business processes
  - Due to the extensive activities, teamwork will be encouraged *(but not enforced!)*
  - ***Real development needed!***

Practicals (Spring Term – TB2)

- Timetabled in 5 Sessions:
  - In room **3Q44**
- A series of development activities are planned for practical sessions
  - **Django/Django Rest Framework** is chosen to use as development framework
  - **Docker** is to be practiced for containerisation and deployment
  - **You're the development team in need** → ***Teamwork will be practiced and assessed accordingly***

Practicals

- Tools to be used are:-
  - ***MS Teams:*** The working teams are expected to meet up and do practical work collectively using this tool.
  - ***BB Groups:*** Each group is allocated a space on BB to use, for communication, exchanging data etc.
  - ***GitHub/GitLab:*** The worked out and developed code is expected to be stored in GitHub repos and to be made available to review.
  - A professional project management (e.g., Jira, Microsoft Project, Trello, etc) tool of your choice should be used.
  - Others,… *based on choice again.*

Groups

- Group-based work practices starts in TB1 and will continue throughout the module.
- **Sprint-based development** is a standard practice applied throughout the year across 2 assessment tasks.
- Group works will be assessed.
- **Individual contributions matter!!!**

Assessment

- There are mainly 2 assessments:
  - **Task 1**: Group Project – BPMN Modelling
    - Weighing **40%**
    - Modelling & implementing executable BPMN solution
    - Due on the last week of Autumn Term – TB1
    - **Final In-class Demos will be scheduled**
  - **Task 2**: Group Project – System Development:
    - Weighing **60%**
    - Development of the system in Django/Django Rest
    - Evaluate against provided test cases
    - Due on last week of Spring Term – TB2
    - **Final In-class Demos will be scheduled**

Task 1 – Group Project – BPMN

- **Assessment Task 1** is a group project with **3 sprints**.
  - You will need to create three interconnected models:
    - A strategic-level BPMN diagram representing enterprise system architecture
    - An operational-level BPMN diagram detailing specific components and services
    - An i* socio-technical model capturing user goals and dependencies
  - Provide critical evaluation of the system with particular emphasis on discussing legal, ethical, social, and professional implications of the processes and interactions represented in your models
  - BPMN operational models need to be executable. They will be treated as micro-services for the system development at the later stage.

Task 2 – Group Project – Django

- **Assessment Task 2** is a group project with **3 sprints**.
  - **Django / Django Rest Framework**, **Docker**, **and corresponding Database(s)** must be applied for system development
  - **Solid understanding** of the developed is **required**. *Demonstrating a working system is insufficient!* ***YOU KNOW WHY!***
  - The group effort will be assessed. You need to use ***version control software, GitHub/GitLab,*** to record your team-works/commits.
    - ***Kanban*** facility of GitHub is also useful for project management.

Group Work

- It is essential to start teamwork with your group members **(4-5 members)**.
  - You can choose your group members from other practical sessions.
  - *Group formation* ***starts from now*** *until* ***07th October 2026***.
  - Students with no groups will be randomised into groups.
- ***Once formalised, groups remain unchanged throughout assessments Task 1 and Task 2.***
- You are expected to *work together* with your group members during *practical sessions* especially in 2nd hour of the practical sessions.
- Weekly **Stand-ups** are assessed individually and will contribute to individual marks.
- At the end of each Sprint, a **Sprint Review** will take place. A 10-minute demonstrating the project progress is required. *Format: video/in-person – your choice.*
- **Again, individual contributions matter!**

Group Formation Guidelines – **Requirements & Deadlines**

- **Size:** 4–5 members per group (maximum 5).
- **Deadline:** You may self-select your group until **23:59 (midnight) on 7th October 2026.**
- **Random Allocation:** After this date, any student without a group will be randomly assigned to a new group or added to an existing group with fewer than 5 members.
- **Lock Date:** Group membership is final once Sprint 1 begins on **12th October 2026**.

Group Formation Guidelines – **Cross-Session Logistics**

- **Formation:** Groups may be formed across different practical sessions. However, *please try your best to form groups within your allocated timetabled sessions*. We will *prioritise group members from the same timetabled session to avoid timetable disruption*.

- **Scheduling:** As groups are required to attend weekly standups with the tutor, members should *ideally request a timetable change to align their sessions*.
  - If a formal change is not possible, the group must agree on a single common session to attend **consistently throughout the terms** and notify the tutor.

- **Attendance Code:** Please note that *attendance codes are strictly linked to your official timetable sessions* and will not function for sessions you are not registered to attend.

Group Formation Form



DESD Group Formation 2026-27 Form

Feedback

- Formative Feedback:
  - We plan to provide feedback to group works during practical sessions. Please ask for feedback if you feel you need more details on practical works.
  - Feedback is not only about your mistakes or areas to be improved but also about appreciation of your achievements.
- Summative Feedback:
  - Provided once you have submitted your assessment materials. This includes the marks decided for the work with strengths and weaknesses, which justify the mark decided.